Localisation in the paired-cut equality case

Human Compression project: L044 working continuation

Use the notation and square-normal-form theorem of *Paired orbit cuts in string actions*. Suppose $2r = n$. Among the paired selections whose only cycles are alternating squares let Q minimise the number q of squares.

Lemma 1. *Every such minimum-square selection has exactly one square in every component.*

Proof. If some minimum-square selection had two squares in one component, choose, over all such selections, a pair at the smallest distance. The incident-square, ordinary-transport and global middle-colour moves in the square-normal-form proof contradict first the minimal square count and then this minimal distance. Thus every component has at most one square. The exact defect $2r = n - c + q$ and $2r = n$ give $q = c$, so there are no tree components. □

Lemma 2 (Localising cycle colours)**.** *Let S be the square in a component C, and let a be one of its two colours. Every eligible a-edge of the full permutation graph has both endpoints in C.*

Proof. If both endpoints lay in another component, its connecting path in Q would avoid a, contradicting eligibility. If the endpoints lay in different components, delete one selected a-edge of S and insert the new edge. The deletion destroys S without disconnecting C, while the new edge joins different components. No cycle is created, so the number of squares decreases. This also is impossible. The only remaining location is within C. □

Lemma 3 (Full neighbours of square vertices)**.** *Every edge of the full permutation graph incident with a vertex of S has its other endpoint in C.*

Proof. Let the square colours be a, b, and let $v \in S$. The cases of colours a, b are internal to S. If a colour k commutes with one of them, say a, then the image of an a-edge under $\rho_k \in G_a$ is an actual eligible a-edge. Lemma 2 places both its endpoints, including v^{ρ_k}, in C.

The remaining case has $a = i - 1, b = i + 1, k = i$. Use the global middle-colour exchange, obtaining Q' with the same minimum q. Viewed as a graph operation on Q^{ρ_i}, this removes S^{ρ_i} and inserts S. Removing the four edges of S^{ρ_i} splits C^{ρ_i} into four trees; every other component remains unicyclic. All four vertices of the inserted S must lie within C^{ρ_i}. Otherwise the component containing that vertex in Q' would contain both the new square S and the unchanged old square of another component, contrary to Lemma 1. Therefore $S \subseteq C^{\rho_i}$, equivalently $S^{\rho_i} \subseteq C$. In particular $v^{\rho_i} \in C$. □

Lemma 4 (Ordinary slides preserve the component partition)**.** *Let a selected bridge $v -_\ell w$ leave S, with ℓ commuting with a square colour a. The ordinary square slide may be made without changing any component's vertex set.*

Proof. Write $v' = v^{\rho_a}$ and $w' = w^{\rho_a} = (v')^{\rho_\ell}$. By Lemma 2, the eligible new edge ww' lies within C. Delete the opposite a-edge of S, add ww', and complete the parallel ℓ-edge $v'w'$ if necessary, deleting the other selected ℓ-edge. As in ordinary transport, the number of squares cannot increase and no other cycles are created; minimality forces the square count to stay q.

If the removed ℓ-edge were outside C, it would be a bridge: every cycle containing it would also require the selected ℓ-edge vw, which lies in C. Removing that outside bridge leaves a tree component, and all inserted edges are within C. This contradicts Lemma 1. Thus every changed edge lies in C. Its new selection has one square and cannot leave a tree component, again by Lemma 1; hence it remains connected. Every other component is untouched. □

Lemma 5 (The transported partition)**.** *For a square S of colours $i-1, i+1$, the global middle-colour exchange has component partition exactly the image under ρ_i of the old component partition.*

Proof. The proof of Lemma 3 shows that every vertex of the inserted S lies in C^{ρ_i}. Deleting S^{ρ_i} split this component into four trees. Inserting S must reconnect all four: any omitted piece would remain a tree, contrary to Lemma 1. All other components are the untouched images of the old components. Thus the new component containing S has exactly the vertex set C^{ρ_i}, as claimed. The claim is transport of the partition, not invariance of the old partition. □

Lemma 6 (Connected equality normal form)**.** *If $2r = n$, a minimum-square paired selection is connected and has exactly one square.*

Proof. The minimum number q of squares is fixed. If a minimum-square selection is connected then Lemma 1 already gives $q = 1$; the same is then true of every minimum-square selection. Suppose instead that all such selections are disconnected. The full permutation graph is connected by transitivity. Hence for any of these selections there is a full graph edge between two components, say C, D, and there are selected paths from their unique squares S, T to the respective endpoints. Choose, over *all* minimum-square selections and *all* such edges and selected tails, a connector P with the fewest edges. Its only unselected edge is this cross-edge, whose colour will be denoted by x.

Both selected tails are nonempty, by Lemma 3. Suppose the first edge from S has colour ℓ and commutes with a colour a of S. Apply the ordinary slide, with notation v, v', w, w' as above. If its other selected ℓ-edge lay on P, then w' would lie outside C: on the w side of the bridge vw a path to w' would avoid a, whereas on its other side a path from v' to w' would avoid ℓ. The same argument works whether the other ℓ-edge lies later in the C tail or in the D tail. Both alternatives contradict eligibility, exactly as in the repeated-colour case of ordinary transport. But ww' is an eligible a-edge, so Lemma 2 requires $w' \in C$. Thus the other ℓ-edge is not on P.

Consequently the ordinary slide preserves the remaining connector. It also preserves its two component vertex sets by the preceding slide lemma. Its new square includes w, shortening the selected tail by one, contrary to the choice of P. Thus the first edge at S must have a middle colour i with square colours $i-1, i+1$. The same argument at T gives a middle colour j and square colours $j-1, j+1$.

The two square-colour sets are disjoint. In particular the cross-edge colour x lies outside at least one of them; choose that endpoint, say S. Then ρ_i commutes with ρ_x, including when $x = i$. Perform the global middle-colour exchange at S. Every selected edge of the connector other than the first is outside S, so is transported to a selected edge. The single full cross-edge is transported to an actual x-edge because of this commutation. By Lemma 5 it still crosses exactly the two transported components. The transported tail begins at $w^{\rho_i} = v \in S$ and ends at the transported T, and has one less edge. It is therefore an admissible shorter connector, the final contradiction. □

Attribution and remaining step. The source's Proposition 4.12 gives the relevant disconnected reduction using three local connector configurations. The proof above instead combines the root-supplied global middle-colour move with minimum-square localisation. It makes no new source, priority or conjecture claim. The connected equality selection still needs the source's rigidity argument, giving a preserved system of pairs for $n \geq 9$ and its exact two ladder groups. Connectedness alone does not yet exclude the alternating action.